Roberto **Stanzione**

PostDoc · École normale supérieure

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Summary and Research Interests

I received my Ph.D. at the *University of Salerno*, where my research project focused on **data profiling** and the definition of **metadata-driven approaches** for data management tasks. My work focused on a specific type of multi-attribute constraint, i.e., **Relaxed Functional Dependencies (RFDs)**. My research examined the use of RFDs for **data imputation**, **query relaxation**, and **concept drift detection**. I am currently a postdoc researcher at *École Normale Supérieure*, where my research focuses on **time series analysis** and the application of **foundation models and large language models (LLMs)** to temporal data. The latter includes the study of **time series representations** for effectively providing temporal information to foundation models. I am also interested in **time series anomaly detection**, **segmentation**, and **motif discovery**, with a particular focus on understanding the capabilities and limitations of traditional methods, foundation models, and human experts across these tasks. Furthermore, my research explores the use of LLMs to support the interpretation of anomalies in time series. I also continue to investigate **data quality** in time series and its implications for downstream analysis.

University Experience

École normale supérieure
Paris, France

PostDoc Researcher — *Jan. 2026 - Present*

Supervisor: Paul Boniol.

University of Salerno
Salerno, Italy

Reserch Fellow — *Nov. 2025 - Dec. 2025*

Supervisor: Loredana Caruccio.

Visiting Positions

École normale supérieure
Paris, France

Ph.D. Visiting Period — *Feb. 2025 - Apr. 2025*

Supervisor: Paul Boniol.

Université Paris Cité
Paris, France

Ph.D. Visiting Period — *Nov. 2024 - Jan. 2025*

Supervisor: Themis Palpanas.

Education

University of Salerno
Salerno, Italy

Ph.D. in Computer Science — *Nov. 2022 - Nov. 2025*

Research Topic: Metadata-Driven Data Management and Analysis: from Profiling to Applications.
Advisor: Loredana Caruccio.
Final Grade: Excellent.

University of Salerno
Salerno, Italy

Master's Degree in Computer Science. — *Sep. 2019 - Apr. 2022*

Thesis Title: Data Imputation and RFD Discovery in Incremental Scenarios: A Discovery Approach Supporting Imputation
Final Grade: 110/110 with honors.

University of Salerno
Salerno, Italy

Bachelor's Degree in Computer Science. — *Sep. 2015 - Sep. 2019*

Thesis Title: Generation, Delivery, and Reporting of Tests for Supporting Dyscalculia Detection
Final Grade: 102/110.

Publications

2026 **NeurIPS 2026***, *Retrieval Over Training: Similarity search-based Model Selection for Time Series Anomaly Detection. (Panourgias, C., Stanzione, R., Petralia, A., Palpanas, T., & Boniol, P.)*

2026 **NeurIPS 2026***, *Detect, Explain, Interpret: An End-to-End Benchmark for Time Series Anomaly Detection, Explainability and Interpretability. (Stanzione, R., Barbe, J., Parrino, M., Fourmann, J., & Boniol, P.)*

2025 **CRISTAIN 2025**, *Grace: Graph-based Representation and Analysis for Crime Exploration. (Iuliano, G., Margarella, G., Polese, G., & Stanzione, R.)*

2025 **SEBD 2025**, *Concept Drift Detection in Machine Learning Systems by Exploiting Relaxed Functional Dependencies. (Caruccio, L., Cirillo, S., Polese, G., & Stanzione, R.)*

2025 **EDBT 2025**, *An RFD-Based Approach for Concept Drift Detection in Machine Learning Systems. (Caruccio, L., Cirillo, S., Polese, G., & Stanzione, R.)*

2024 **IEEE BigData 2024**, *RYAN: A tool for explaining and visually analyzing the evolution of Relaxed Functional Dependencies. (Caruccio, L., Cirillo, S., Iuliano, G., Polese, G., & Stanzione, R.)*

2023 **MTAP**, *Surveying federated learning approaches through a multi-criteria categorization. (Caruccio, L., Cimino, G., Deufemia, V., Iuliano, G., & Stanzione, R.)*

2023 **EDBT 2023**, *REQUIRED: A Tool to Relax Queries through Relaxed Functional Dependencies. (Caruccio, L., Cirillo, S., Deufemia, V., Polese, G., & Stanzione, R.)*

2022 **ITADATA 2022**, *Data Analytics on Twitter for Evaluating Women Inclusion and Safety in Modern Society. (Caruccio, L., Cirillo, S., Deufemia, V., Polese, G., & Stanzione, R.)*

*Accepted, to appear soon.

Program Committees

2026 **VLDB 2026**, 52nd International Conference on Very Large Data Bases

2026 **DSIT 2026**, 9th International Conference on Data Science and Information Technology

2025 **RAISE 2025**, 1st RAISE Workshop: Resilient AI Framework for Secure and Ethical Machine Learning Systems

Organizing Committees

2026 **ICWE 2026**, 26th International Conference of Web Engineering (Role: **Proceedings chair**)

Reviewer (Conferences)

2026 **DSIT 2026**, 9th International Conference on Data Science and Information Technology

2026 **SHIELD-AI 2026**, Security, Human Awareness, and Risk Mitigation in AI-driven Systems (Workshop at ECML PKDD 2026)

2026 **BDA 2026**, Data Management Principles, Technologies and Applications (Demonstration track)

2026 **ECML PKDD 2026**, European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (Workshop Track)

2026 **ICDM 2026**, International Conference on Data Mining 2026

2026 **NeurIPS 2026**, The Fortieth Annual Conference on Neural Information Processing Systems (Research Track)

2026 **KDD 2026**, ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (Dataset & Benchmark Track)

2026 **KDD 2026**, ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (Research Track)

2026 **VLDB 2026**, 52nd International Conference on Very Large Data Bases (Demo Track)

2025 **RAISE 2025**, Resilient AI Framework for Secure and Ethical Machine Learning Systems

2025 **ITADATA 2025**, 4th Italian Conference on Big Data and Data Science

2025 **ICDE 2025**, International Conference on Data Engineering

2025 **SEBD 2025**, Symposium On Advanced Database Systems

2024 **EuroUSEC24**, European Symposium on Usable Security

2024 **BDS 2024**, IEEE BigDataService 2024

2023 **BDS 2023**, IEEE International Conference on Big Data Computing Service and Machine Learning Applications

Reviewer (Journals)

DKE, Data & Knowledge Engineering
DSEJ, Data Science and Engineering Journal
DSM, Data Science and Management
ESWA, Expert Systems With Applications
IJIS, International Journal of Intelligent Systems
IJSEKE, International Journal on Software Engineering and Knowledge
IoT-J, IEEE Internet of Things Journal
JDIQ, Journal of Data and Information Quality
MTAP, Multimedia Tools and Applications
SOCO, Soft Computing

Conference Participation

CHITALY 2025 - 16th Biannual Conference of the Italian SIGCHI Chapter — *Salerno, Italy*
Student Volunteer & Research Paper Speaker at the co-located CRISTAIN workshop — *6th - 10th October 2025*

SEBD 2025 - 33rd Symposium On Advanced Database Systems — *Ischia, Italy*
Research Paper Speaker — *7th - 11th June 2025*

EDBT 2025 - 28th International Conference on Extending Database Technology — *Barcelona, Spain*
Research Paper Speaker — *25th - 28th March 2025*

EDBT 2024 - 27th International Conference on Extending Database Technology — *Paestum, Italy*
Student Volunteer — *25th - 28th March 2024*

IEEE BigData 2023 - International Conference on Big Data — *Sorrento, Italy*
Student Volunteer — *15th - 18th December 2023*

EDBT 2023 - 26th International Conference on Extending Database Technology — *Ioannina, Greece*
Research Demo Paper Speaker — *28th - 31st March 2023*

Invited Talks

Transfer Learning and Time Series Foundation Models — *Palaiseau, France*
Retrieval Over Training: Similarity search-based model selection for Time Series Anomaly Detection — *3th - 4th November 2026*

Teaching

2025 **PSL Université Paris**, Lab Instructor (Tutorials) for the ”Databases” course (22.5 hours) — *Paris, France*
2025 **University of Salerno**, Teaching Assistant for the ”Database Systems” course (20 hours) — *Fisciano, Italy*
2024 **University of Salerno**, Teaching Assistant for the ”Database Systems” course (20 hours) — *Fisciano, Italy*
2024 **G. Marconi Higher Education Institute**, Tutor for the ”Social Robotics in the Arts” project (20 hours) — *Nocera, Italy*
2023 **University of Salerno**, Teaching Assistant for the ”Database Systems” course (40 hours) — *Fisciano, Italy*